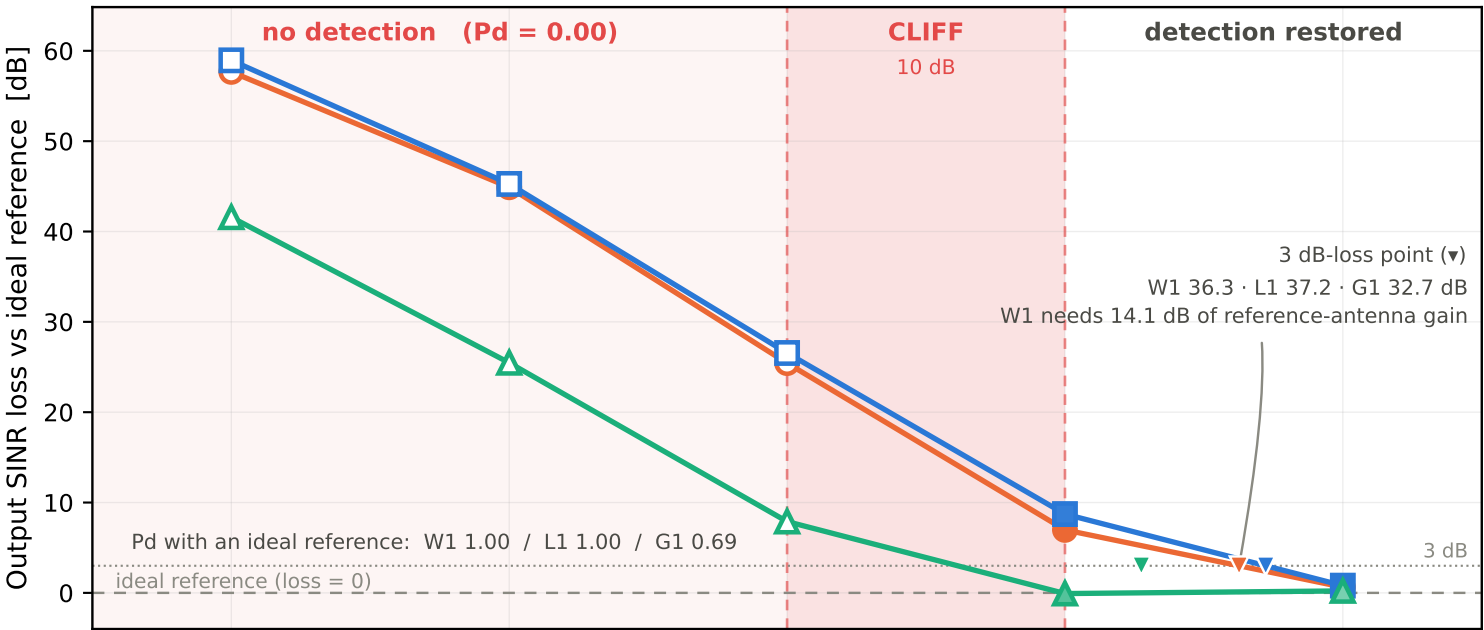


How much does passive detection lose when the reference channel is real?

Noise in the reference channel is a cliff — multipath in it is flat ground.

W1 · WiFi VHT-LTF (76.6 MHz ref) L1 · LTE CRS (18.0 MHz ref) G1 · 5G NR SSB (7.2 MHz ref)

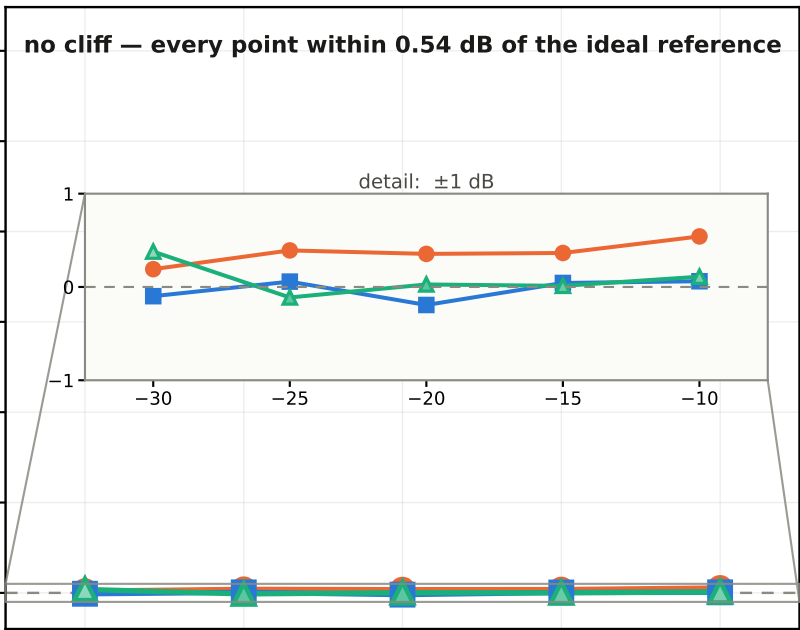
(a) reference-channel noise



W1	0.00	0.00	0.00	1.00	1.00
L1	0.00	0.00	0.00	0.96	1.00
G1	0.00	0.00	0.00	0.75	0.59
	0	10	20	30	40

Reference-channel SNR [dB]

(b) reference-channel multipath — same vertical scale



W1	1.00	1.00	1.00	1.00	1.00
L1	1.00	1.00	1.00	1.00	1.00
G1	0.50	0.72	0.72	0.66	0.59
	-30	-25	-20	-15	-10

Multipath-to-direct ratio in the reference channel [dB]